AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M251
Object Oriented Programming Using Java

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15 / Nov. / 2025

AOU EGYPT

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

1. This exam consists of three parts. It is required to solve **ALL** of them in the external answer booklet:
   - **Part 1: 5** MCQ (10 marks)
   - **Part 2: 2** short answer questions (20 marks)
   - **Part 3: 3** problem solving questions (30 marks)
2. Go through all questions before you start answering them.
3. It is not allowed to use any external material.
4. It is not allowed to use calculators or any electronic device.
5. At the end of the examination:
   - Check that you have written **your name, student ID and section number** on the first page of your answer booklet.
   - Place your question paper in the answer booklet and hand over them to the invigilator.

CamScanner

PART 1: MCQ [10 Marks: 5x2]

Question 1: [10 marks] Read the following questions and choose the best answer.

1. The Java compiler converts _____ to bytecode.
   a. source code b. machine code c. intermediate code
   d. native code e. None of the above

2. The default value of local variables declared as int is _____.
   a. -1 b. null c. '\u0000'
   d. 0 e. There is no default value for the local variables

3. What is the output of the following code if `s = "Toshiba"` ?
   `System.out.println(s.substring(4));`
   a. h b. i c. Tosh
   d. iba e. hiba

4. The value of variables declared as _____ cannot be changed after initialization.
   a. last b. constant c. private
   d. final e. protected

5. The _____ of an ArrayList object is the current number of items stored in the ArrayList.
   a. capacity b. size c. strength
   d. All the above e. None of the above

AOU EGYPT

PART 2: Short Answer Questions [20 Marks]

Question 2: [10 marks: 2+2+(3x2)]
Consider the following code and then answer the questions below it.

```
public class Mobile {
  private double sizeInInches;
  public Mobile (double s){
    sizeInInches = s;
  }
  public double calSizeInCM(){
    return 2.54*sizeInInches;
  }
}
```

```
public class Test {
  public static void main(String[] args) {
    Mobile m1 = new Mobile(); // line 1
    m1.sizeInInches = 6.2;    // line 2
    // code lines
  }
}
```

A. In the class **Test**, is the code in line 1 **valid**, or will cause a compilation error? **Explain**.
B. In the class **Test**, is the code in line 2 **valid**, or will cause a compilation error? **Explain**.
C. In the class **Test**, replace the comment "`// code lines`" with code implementing the following requirements:
   1. Declare and create an array of **Mobile** objects where the array size is 2
   2. Create two **Mobile** objects and assign them to the two elements of the array
   3. Using for loop, iterate over all **Mobile** objects of the array and print their size in **CM**

CamScanner

Question 3: [10 marks: 5x2]

Write the missing code which do the following:

1. Declare and create an **array list** called **brands** which stores brand names.
2. Insert the following values into the list: Sony, Usha, Toshiba, Orient.
3. Display on the screen the first value in the list by calling a built-in method.
4. Sort the list by calling a built-in method.
5. Display on the screen the contents of the list after sorting by calling a suitable method.

```
import java.util.*;
public class CompleteCode {
    public static void main(String[] args) {
        //missing code
    }
}
```

PART 3: Problem Solving Questions [30 Marks]

Question 4: [10 marks: 1+3+4+2]

Write the class **Machine** which includes: (follow **OOP principles** you studied in the class)

1. two attributes to store **the brand** (e.g., Usha) and the **price** (e.g., 16.5) of the **Machine.**
2. a zero-arg constructor and a multi-arg constructor.
3. getters and setters for the two attributes.
4. **toString()** method which overrides the Object's toString() method.

Question 5: [13 marks: 2+3+1+2+5]

Write the class **Fan** that inherits the class **Machine**, and includes the following:

1. one attribute to store the **size** (in inches) of the **Fan**. (E.g., 36)
2. a mutli-arg constructor which invokes the parent class constructor.
3. getter for the attribute **size**.
4. **toString()** method which invokes toString() method of the parent class.
5. **equals()** method which overrides the Object's equals() method. All attributes (including the inherited ones) should be compared.

Question 6: [7 marks: 2+2+(3x1)]

Write a Test class to test your classes by performing the following tasks:

1. Declare, create 2 **Fan** objects, and initialize them by valid values.
2. Print a message explaining if the 2 **Fan** objects are identical or not.
3. Display on the screen the **size** of the first **Fan** by calling a suitable method.
4. Change the **price** of the first **Fan** to **17.25**
5. Display on the screen all information of the second **Fan** by calling no more than one method.

End of questions

TOLEGKPT

CamScanner